Developmental: Classic Label Propagation and Active Learning on Graphs

Overview

In settings where labeling data is expensive, graph-based semi-supervised learning provides powerful tools for propagating information from a small labeled subset to unlabeled nodes. The classical Gaussian Fields and Harmonic Functions (GFHF) method of Zhu et al. [4] models labels as a smooth function over the graph, enabling inference through iterative relaxation.

This project invites students to implement classical label propagation and extend it with an active learning component, selecting nodes that maximize expected uncertainty reduction or structural influence. Students will simulate labeling scenarios, incorporate simple noise models, and evaluate how query strategies influence accuracy and robustness.

Expected outcomes include implementations of label propagation, heuristic or uncertainty-based query selection, and experiments comparing labeling strategies under varying budget constraints. Students may also explore connections to crowdsourcing noise models as described by Fang et al. [2] and Zhang et al. [3].

Related Work

The GFHF algorithm was introduced by Zhu et al. [4], forming a core technique in classical graph-based semi-supervised learning. Active learning for networked data was studied by Bilgic [1], and extensions to noisy crowdsourcing scenarios appear in Fang et al. [2] and Zhang et al. [3].

References

[1] M. Bilgic. Active learning for networked data. In *Proceedings of the 27th international conference on machine learning (ICML-10)*, 2010.

[2] M. Fang, J. Yin, and D. Tao. Active learning for crowdsourcing using knowledge transfer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 28, 2014.

[3] W. Zhang, Y. Wang, Z. You, Y. Li, G. Cao, Z. Yang, and B. Cui. Nc-alg: Graph-based active learning under noisy crowd. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, pages 2681–2694. IEEE, 2024.

[4] X. Zhu, Z. Ghahramani, and J. D. Lafferty. Semi-supervised learning using gaussian fields and harmonic functions. In *Proceedings of the 20th International conference on Machine learning (ICML-03)*, pages 912–919, 2003.